THE DECOMPOSITION OF THE PUSHFORWARD OF THE INTERSECTION COHOMOLOGY SHEAF OF THE REGULAR PART OF A QUIVER VARIETY

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Allowed standard prerequisites (editorial).

- The local systems phi on the strata of M_0(w) and the decomposition of the top-degree homology of the fibre as a direct sum of phi tensor V_(G-hat),phi. Established in the previous subsection and used at the start of the counted proof (proof lines 2734-2738). Supplies the g-module decomposition whose summands the rest of the proof identifies.
- The intersection cohomology sheaves IC of the strata and their behaviour under a finite birational morphism, in particular the projection formula used for kappa and kappa-prime. Used inside the counted proof at source lines 2766-2776. Justifies replacing the intersection cohomology of the closure of a product stratum by the pushforward of the exterior product under the finite birational morphism.
- The geometric description of the regular strata: on the preimage of a regular stratum the map pi_{0,zeta} is an isomorphism, and the top-degree

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homology of the fibre is the irreducible integrable highest weight representation with highest weight vector the class of x. Attributed by the author to section 10.2 of the cited paper Na-alg and used at proof lines 2742-2749. Identifies the two families of summands appearing in the final formula: the trivial-local-system case on the regular strata of M_0 and the irreducible representation V^(g_I^0 + h)(w - v^0) on the regular strata of M_{zeta^bullet}.

Source reference. Original equation 5.2: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 2625–2625. Navigation pointer only: it labels the author's displayed decomposition of the pushforward that the counted proof refers to when it defines the multiplicities; the display itself is not reproduced.

Source reference. Original equation 5.3: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 2634–2634. Navigation pointer only: it labels a second displayed relation of the author that the proof substitutes into; the display is not reproduced.

Source reference. Original equation 2.17: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1658–1658. Navigation pointer only: it labels the author's displayed product description of a stratum, which the counted proof recalls in one clause; the display is delivered verbatim inside the statement span of the card for prop:Levi but is not part of this card.

Source reference. Original equation 2.18: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1738–1738. Navigation pointer only: it labels the author's commutative diagram used in the middle step; the diagram is not reproduced.

Theorem 0.1. *Let $V^{\mathfrak{g}}(\lambda)$ (resp. $V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\lambda)$) denote the irreducible integrable highest weight module of $\mathfrak{g}$ (resp. $\mathfrak{g}_{I^0}+\mathfrak{h}$) with the highest weight λ. Then we have*

$$(\pi_{0,\zeta^\bullet})_*(IC(\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}))) = \bigoplus_{\mathbf{v}'} \operatorname{Hom}_{\mathfrak{g}_{I^0}+\mathfrak{h}}(V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\mathbf{w}-\mathbf{v}^0), V^{\mathfrak{g}}(\mathbf{w}-\mathbf{v}')) \otimes IC(\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w})) \\ \oplus \bigoplus (IC \text{ sheaves associated with } \textit{non}\text{-regular strata}).$$

Proof. Let $\mathfrak{M}_0(\mathbf{w})_{(\widehat{G})}$ be a stratum of $\mathfrak{M}_0(\mathbf{w})$. Take a point $x \in \mathfrak{M}_0(\mathbf{w})_{(\widehat{G})}$. The decomposition $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x)) = \bigoplus \phi \otimes V_{(\widehat{G}),\phi}$ in the previous subsection is a decomposition of a $\mathfrak{g}$-module. Suppose $\mathfrak{M}_0(\mathbf{w})_{(\widehat{G})}$ is regular, i.e., $= \mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w})$ for some $\mathbf{v}'$. Then it is known that $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x))$ is an irreducible integrable representation of $\mathfrak{g}$ with the highest weight vector $[x] \in H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x) \cap \mathfrak{M}_\zeta(\mathbf{v}',\mathbf{w}))$ [6, 10.2], where x can be considered as a point in $\mathfrak{M}_\zeta(\mathbf{v}',\mathbf{w})$ as $\pi_{0,\zeta}\colon \mathfrak{M}_\zeta(\mathbf{v}',\mathbf{w}) \to \mathfrak{M}_0(\mathbf{v}',\mathbf{w})$ is an isomorphism on the preimage of $\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w})$. In particular, $V_{(\widehat{G}),\phi} = 0$ unless ϕ is the trivial local system in this case.

Similarly if a stratum $\mathfrak{M}_{\zeta^\bullet}(\mathbf{w})_{(\widehat{H})}$ is of the form $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w})$ for some $\mathbf{v}^0$, then $H_{\mathrm{top}}(\pi_{\zeta^\bullet,\zeta}^{-1}(y))$ is the irreducible integrable representation $V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\mathbf{w}-\mathbf{v}^0)$ if $y \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w})$. The highest weight vector is $[y]$ as above.

Consider a stratum $\mathfrak{M}_{\zeta^\bullet}(\mathbf{w})_{(\widehat{H})}$ which contains $IC(\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w}))$ in the decomposition (5.2). We can write it as $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times (\mathfrak{M}_0)_{(\widehat{H}')}$ as in (2.17). We consider the diagram (2.18). Since κ is a finite birational morphism, we have

$$IC(\overline{\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times (\mathfrak{M}_0)_{(\widehat{H}')}}) = \kappa_*\big(IC(\mathfrak{M}(\mathbf{v}^0,\mathbf{w})) \boxtimes IC(\overline{(\mathfrak{M}_0)_{(\widehat{H}')}})\big),$$

and hence

$$(\pi_{0,\zeta^\bullet})_* IC(\overline{\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times (\mathfrak{M}_0)_{(\widehat{H}')}}) = \kappa'_*\big((\pi_{0,\zeta^\bullet})_* IC(\mathfrak{M}(\mathbf{v}^0,\mathbf{w})) \boxtimes IC(\overline{(\mathfrak{M}_0)_{(\widehat{H}')}})\big).$$

Since κ' is also a finite morphism, this contains only IC sheaves on nonregular strata unless the second factor $(\mathfrak{M}_0)_{(\widehat{H}')}$ is trivial. Hence (5.3) now becomes

$$\operatorname{Res} V^{\mathfrak{g}}(\mathbf{w}-\mathbf{v}') = \bigoplus_{\mathbf{v}^0} V^{\mathbf{w}-\mathbf{v}^0}_{\mathbf{w}-\mathbf{v}'} \otimes V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\mathbf{w}-\mathbf{v}^0), \tag{0.2}$$

where $V^{\mathbf{w}-\mathbf{v}^0}_{\mathbf{w}-\mathbf{v}'}$ is the multiplicity of $IC(\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w}))$ in the decomposition of $(\pi_{0,\zeta^\bullet})_*$ $(IC(\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w})))$ in (5.2). Thus $V^{\mathbf{w}-\mathbf{v}^0}_{\mathbf{w}-\mathbf{v}'} = \operatorname{Hom}_{\mathfrak{g}_{I^0}+\mathfrak{h}}(V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\mathbf{w}-\mathbf{v}^0), V^{\mathfrak{g}}(\mathbf{w}-\mathbf{v}'))$. Note here that $V^{(\mathfrak{g}_{I^0}+\mathfrak{h})}(\mathbf{w}-\mathbf{v}^0)$ are non-isomorphic for different $\mathbf{v}^0$. □

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